

AI-powered approaches to solve real-world scientific problems

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BY

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to the

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KALINDI COLLEGE
(UNIVERSITY OF DELHI)

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1. Details of project proposal

- Subject Area of the Project - Artificial Intelligence
- Title of the Project - AI-powered approaches to solve real-world scientific problems.

2. Details of all Principal Investigator

1. Dr. Savita Sharma
2. Dr. V. Bhasker Raj
3. Dr. Reena Jain

3. Names, Roll numbers and Course details of the Students

Sr. No.	Names	Roll no.	Course
1.	Afreen Chaudhary	23567026	BSc.(H) Physics III Year
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4. Project progress details

(i) Summary of work done :-

- a. Understanding the Basic principles of E-nose systems and their applications in gas sensing and pattern recognition.

- b. Structural format of sensor response data corresponding of different analytes was collected and studied.
- c. To reduce dimensionality, analyze variance distribution and check redundancy among sensor signals, PCA was applied to the data.
- d. Designing and Training of Artificial neural network (ANN) using raw sensor data.
- e. Improvement of the classification performance and generalization capability by using PCA output features as input for ANN.
- f. Results were in the form of tables and plots.
- g. **Conference attended/ Paper presented/ published ---**

- 1) "Role of AI in enhancing Research Methodology", **Afreen Chaudhary, Shweta Sharma, V. Bhasker Raj, Savita Sharma, Reena Jain**, presented at National conference on "Research Methodology in the Digital Age: Innovations, Ethics, and Applications" organized by Research Committee of Kalindi College held at Kalindi college, University of Delhi.
- 2) "AI-assisted automation and optimization of physics experiments", **Shweta Sharma, Afreen Chaudhary, V. Bhasker Raj, Savita Sharma, Reena Jain**, presented at 12th IAPT National Student Symposium on Physics (NSSP 2025) held at Panjab University, Chandigarh.
- 3) Book Chapter 8- AI-Powered Learning Analytics and Decision-making in Higher Education in book "Education 4.0: Class to Career", **V. Bhasker Raj, Afreen Chaudhary, Shweta Sharma, Savita Sharma**, published by Sahitya Sansthan (2025). pp.110–127. ISBN: 978-93-95397-69-B.
- 4) Book Chapter 9- AI-Powered Learning Analytics and Decision-making in Higher Education in book "Education 4.0: Class to Career", **Savita Sharma, Madhavi Sharma, Saniya, V. Bhasker Raj** published by Sahitya Sansthan (2025). pp.128–143. ISBN: 978-93-95397-69-B.

Project Aim: AI-powered approaches to solve real-world scientific problems.

1. Introduction

The main goal of this project is to use intelligence to solve real world problems that scientists are facing. Environmental pollution is one of the main problems these days [7]. It is affecting people's health and the environment in a bad way [7]. People have difficulty identifying how to measure the presence of hazardous elements because they tend not to be able to easily use instruments they should have access to. This is a major concern related to pollution and why the study will investigate alternative ways of addressing this issue. Therefore, this project focuses on the application of artificial intelligence techniques such as PCA and Artificial Neural Networks (ANN) for the intelligent detection and classification of pollutants [5], [14].

1.1 Pollution Related Problems: Everyday Problems

One of the most serious challenges faced by today's society is the Environmental Pollution. It has adverse effects on the environment and significantly impacts human health and quality of life [4]. Various factors like industrialization, increase in urbanization and lack of waste disposal facilities have led to contamination of air, water, soil and overall environment.

A severe threat to human health is continuously inhaling the harmful gases and toxic chemicals present in the polluted air. Vehicles, research laboratories, industrial emission are the major contributors to the air pollution [5]. Its long exposure can cause respiratory diseases such as asthma, coughing, lung cancer and breathing problems. Apart from health issues it also causes environmental degradation. The primary impact of this disturbance causes negative effects on both the psychological well-being of an individual and the entire life experience. It is hence relevant to consider various preventive measures of pollution among different strategies of combating environmental pollution.

1.2 Government-Based Pollution Detection and Monitoring

The Government ensures clean air and water by monitoring the environmental conditions and implementing proper regulation and guidelines [7]. To identify major pollution sources and take

corrective measures on time it is necessary to establish effective monitoring systems across the country.

Pollution control boards and environmental protection agencies are responsible for the pollution surveillance at the national level. They regularly check the concentration of harmful gases and pollutants in the atmosphere to ensure the pollution levels are under the environmental standards.

For accuracy, government agencies have employed lab-based analytical techniques like gas chromatography, mass spectroscopy and real-time monitoring systems [6]. The pollution detection frameworks that are driven by the governmental agencies focusses mainly on continuous monitoring, regulatory enforcement, data analysis, and technological advancements. The integrated methods enable organizations to identify and assess environmental pollution while implementing control measures which lead to improved environmental management and safety.

1.3 Technologies:

To achieve reliable detection, discrimination, and validation of pollution markers, multiple sensing and analytical technologies were studied [6].

1. Conductometric Sensors: Conductometric sensors determine gases by monitoring tensile properties of conflicts in be arrears to differences in emitter concentration or disclaimer carrier density [1]. Metal oxide materials are usually used for their high sensitivity, thermal stability and relatively simple preparation etc [16]. The sensors are cheap, easy to handle and capable of real-time measurement [11]. They have quick response and recovery time as they could be used to detect small variation in gas concentration [7].

2. SAW (Surface Acoustic Wave) Sensors: Surface Acoustic Wave sensors function based on acoustic waves traveling along a piezoelectric substrate surface [9]. Adsorption of gas molecules at the sensor surface modifies both wave velocity and resonant frequency [9]. SAW sensors have the advantages of good sensitivity, rapid response and well reproducibility. They can sense the lowest gas concentrations, enabling environmental control [7].

3. Gas Chromatography (GC): The analytical method of Gas Chromatography enables scientists to achieve precise identification of all components within a gaseous sample [6]. The method utilizes a carrier gas to transport the sample through a column which enables different substances to travel at different rates [6]. The scientists study substances that evaporate from liquids because GC provides them with a powerful research tool. The technology enables gas detection systems to separate their competing sensor signals [6], [18].

4. Surface Plasmon Resonance (SPR): The method of Surface Plasmon Resonance allows the scientists to obtain very precise measurements, which follow the changes in the refractive index at the boundary between metals and dielectrics. The system utilizes light to generate surface electron excitation by directing it at specific angles [10]. SPR provides scientists with the ability to monitor gas adsorption and desorption processes without using any detection labels. The method enables researchers to observe how sensors interact with their target substances at an exceptional degree of accuracy [10].

1.4 Advantages of SAW-Based Sensor System:

The SAW-based sensor system can really be helpful for getting information about pollution. It does a lot of things that other systems aren't capable of [9].

- Extremely high sensitivity, thus the possibility to detect trace-level pollutant concentrations [9].
- Fast response and recovery times, so that monitoring in real time is possible [9].
- Determination of multiple analytes is possible by using a sensor array configuration [8].
- It is highly compatible with pattern recognition and machine learning techniques [5].
- Compact size, low power consumption, and potential for portable device integration [9].

The good things about SAW sensors make them great for use where other forms of gases can interfere. For example, SAW sensors are very important since they are able to handle interfering gases. This is why people love using SAW sensors because they can employ them in environments. SAW sensors are great for use where you have many interfering gases [9].

1.5 Selectivity Techniques:

When we are discussing gas sensing and pollution detection systems, selectivity becomes a very important aspect. This is because gas sensing and pollution detection systems have to function in the real world, where multiple gases are always present simultaneously. Selectivity is one of the factors that we examine to determine the performance capability of such systems [11].

1.5.1 Pattern Recognition Techniques

We analyzed a lot of data from sensors. We applied machine learning to understand what the data means [5]. We applied techniques such as Principal Component Analysis and Artificial Neural Networks to reduce the data and look for patterns [2], [3]. Principal Component Analysis (PCA) is a technique that reduces a lot of data from sensors to a smaller set of key components [2], [4].

Artificial Neural Networks (ANN) were used to reduce data and determine what we were searching for in the data [3]. The combination of PCA and ANN helped to determine the target pollutants correctly [14], [15]. This was achieved when the environmental conditions and concentrations of the gases were not the same. The Principal Component Analysis and Artificial Neural Network complemented each other by improving the selectivity of the target pollutants depending on their concentrations [14].

1.5.2 Sensor Surface Modification

Sensor surface modification is a technique of coating the active sensing surface of sensor, which enhance the selectivity, sensitivity, and overall performance of gas sensing systems, especially in the environment where multiple gases are present simultaneously. Surface modification involves applying coatings onto the active sensing area of the sensor [12]. We choose these coatings according to their chemical affinity toward the target pollutants. When the coatings are properly designed, they protect the sensing material from environmental degradation, humidity effects, and surface contamination. In this way sensor can perform properly even after repeated use and long-term operation [12].

1.5.3 Electronic Nose (E-nose)

A highly developed sensor known as the E-nose imitates how humans detect and identify various gas or odor types by using sensors to analyze incoming data [1]. The system functions using various gas sensors that produce unique chemical reaction patterns for different types of gases [19]. Methods of pattern recognition are utilized to analyze these responses to determine the particular gases present [8]. The E-nose operates through multiple sensors that produce combined responses to detect gases [5], [17], [19].

The Electronic Nose is made up of three components which include: The first component which is the Sensor Array contains sensors that can detect and identify different gases through their specific reactions [16]. The Data Collection System is responsible for collecting data from all existing sensors [19]. The Data Preprocessing Unit processes data from the data acquisition system to create standardized information which the pattern recognition unit will use [5]. The Pattern Recognition Unit uses machine learning methods which include PCA and ANN to evaluate and classify the electronic nose output [1], [3], [14].

When a gas is introduced, each sensor in the electronic nose responds in a different way [19]. Different machine learning methods can be employed to analyze the gas signature. The gas signature can be analyzed using various machine learning techniques [3], [4].

2.1 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a data analysis method that reduces excess dimensionality by transforming multiple dimensions into lower dimensional spaces and attempting to preserve as much of the variability from the original data set as possible. PCA solves this issue by transforming the linked original data dimensions into distinct unconnected principal component dimensions [2].

The principal component represents all original variables through its linear combination which follows the path of highest data variation. The first principal component captures the maximum possible variance, the second captures the maximum remaining variance orthogonal to the first, and so on. PCA achieves data simplification through its principal component selection process which removes unessential elements while decreasing data noise and making visual representation

easier. The project uses PCA to extract the necessary features from the E-nose sensor data, which will improve the efficiency and effectiveness of the ANN model in classification processes [2], [3], [13], [17].

2.1.1 PCA Algorithm

The full process of the Principal Component Analysis algorithm involves the following steps, which must be followed in the given order.

1. The first step in performing PCA on sensor response data is to create a "data matrix", where the rows of the matrix each represent an individual sample and the columns of the matrix represent each field of the sensor.
2. When all of the above has been done, you would calculate a mean value for all elements in each field, and then perform a normalization of the data to be able to remove a previously-established "mean" value from each feature in your data matrix. As all data has been normalized, all data will have a zero-mean value for each field.
3. The covariance matrix of the zero-mean data is computed to measure how well the features of the sensor are related.
4. Eigenvalues and eigenvectors for this covariance matrix will be obtained.
5. The eigenvalues will be arranged from highest to lowest, and the corresponding eigenvectors will also be chosen based upon their order of eigenvalue.
6. The initial data set will be converted to the principal component dimension by means of the eigenvectors chosen in the last step.

2.2 Artificial Neural Networks (ANNs)

ANNs are computer model that comprises of artificial neurons. They imitates the functioning of biological neurons as close as possible. Researchers developed artificial neural networks as computational models which imitate the brain's structure and operation through their design [3].

The system includes multiple processing units which operate as neurons and connect to each other through their layer-based structure. The input signals pass through the neuron which uses its weighted connections and activation functions to produce output results. Through their training process ANN systems develop the ability to detect intricate nonlinear relationships that link input data to output data [2], [4]. The electronic nose gas sensors generate sensor responses which exhibit nonlinear patterns that overlap each other, which decreases the performance of conventional classification methods. The challenges become solvable through ANN because it demonstrates exceptional pattern recognition abilities along with precise classification skills. The ANN model uses PCA to extract features which enable it to identify different gases through their unique sensor response patterns [1], [2], [6]. The use of PCA-reduced inputs improves learning efficiency and classification accuracy [1].

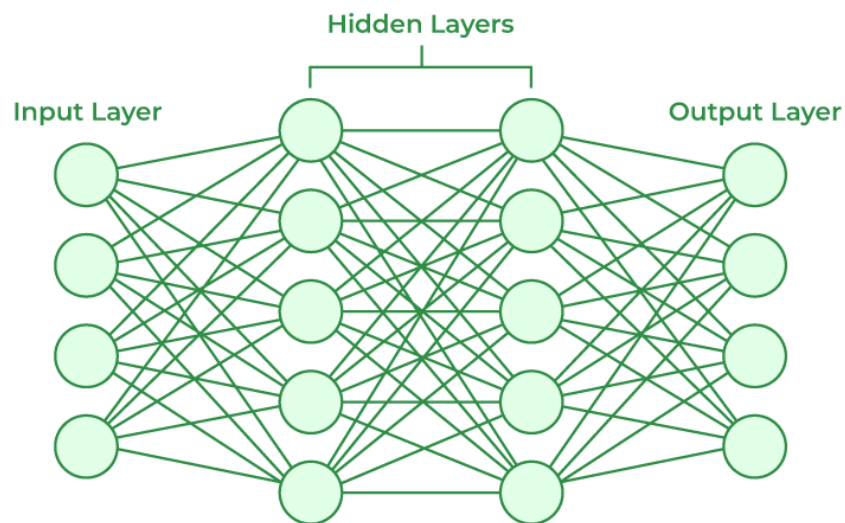


Figure 1 Structure of Artificial Neural Network (ANN)

Input layer: The input layer, after obtaining the data, first forwards it to the network for processing.

Hidden Layers: The system analyzes the input data by creating higher, level features to discover complex patterns.

Output Layer: The system generates the final prediction or classification output.

2.2.1 ANN Algorithm

The following is the algorithm used to implement the artificial neural network:

1. The input layer receives the reduced dimensional feature set that is obtained from PCA.
2. The input signals of every neuron are multiplied by the corresponding weights and then summed up.
3. An activation function is used to introduce non, linearity into the model.
4. The signal is passed on through one or more hidden layers.
5. The output layer generates the expected type of gas.
6. The error is computed by comparing the network's output with the intended output.
7. The back propagation algorithm employs the error to adjust the weights by back propagating the error through the network.
8. The training phase is repeated for a few epochs until the error is at its minimum point.

2.2.2 Types of Artificial Neural Networks

1. **Feedforward Neural Network (FNN):** The Feedforward Neural Network serves as the fundamental neural network model which transmits information from its input to output through a single route that passes through hidden layers.

2. **Convolutional Neural Network (CNN):** Convolutional Neural Networks operate on image grid data to identify important features through their convolutional layers.
3. **Radial Basis Function Network (RBFN):** Radial Basis Function Networks use radial basis functions to perform classification and regression analysis on data which exhibits clear dominant patterns.
4. **Recurrent Neural Network (RNN):** The Recurrent Neural Network processes sequential data and time series data through its feedback connections which function as memory storage.

2.3 PCA–ANN Integrated Model

An electronic nose system that can classify gases using an efficient and accurate method, by combining Principal Component Analysis (PCA) with Artificial Neural Networks (ANN) to form a PCA-ANN model, is used for this study. The PCA method reduces the large number of individual features in a dataset produced by the sensors by finding and removing duplicate or highly correlated features [1]. PCA decreases the dataset dimensions while maintaining all essential information required for gas detection through PCA-ANN feature analysis. The ANN uses the PCA-reduced feature set to train its system which connects each PCA-derived feature with its corresponding gas class through the combination of ANN and PCA methods [1], [2], [19]. The finished ANN model will use sensor output data from the test phase to identify gases in unknown gas samples. The PCA method enhances ANN model performance by enabling use of fewer input neurons which results in faster training time and improved convergence speed and decreased risk of overfitting. The electronic nose system shows better performance and reliability and accuracy results when using PCA and ANN together than when using ANN alone for gas classification [19].

2.3.1 Advantages of PCA-ANN Integrated Model over Individual PCA and ANN Approaches

- PCA diminishes the dimensionality of the sensor data, hence lowering the complexity of the ANN.
- The removal of redundant features enhances the efficiency of learning in the ANN.

- PCA-ANN has fewer input neurons, hence faster training of the ANN.
- The combined model has lower chances of overfitting and improves the generalization of the test data.
- PCA improves the separability of classes, hence increasing the accuracy of classification compared to PCA and ANN separately.
- PCA stabilizes the ANN by reducing noise.

1. Methodology

The methodology adopted for the work involves E-nose based detection and classification of vapors. Four different chemical-sensitive coatings of ZnO, TeO₂, SnO₂, and TiO₂ were applied on to the SAW sensors in the E-nose sensor array for recognizing the individual components in a mixture. The target analytes used in this work includes some fuel samples (Petrol, diesel, kerosene), VOCs (Methanol, Acetone) and simulants (CEPS, DECP, DBS) each of them are measured at different concentration levels.

It is divided into following steps:

3.1 Data Acquisition

E-nose system was used to obtain the dataset used in this project; it consists of 4 gas sensors (Sensor1 - Sensor4). The E-nose was exposed to different categories of vapors, such as fuels, VOCs, and Simulants.

The dataset consists of the following attributes:

- Vapor name
- Target label
- Category
- Gas concentration
- Sensor responses

3.2 Data Pre-processing

To ensure consistency and reliability the sensor data was pre-processed before analysis. From PCA and ANN inputs the non-numerical attributes were excluded. The Sensor response values were standardized to zero mean and unit variance and the dataset was divided into training and testing sets to evaluate the model.

Table 1 Tabular Representation of E-nose Dataset

Vapors	Target label	Category	Concentration	Sensor1	Sensor2	Sensor3	Sensor4
Petroleum	T1	Fuel	1466	207	90	393	464
Petroleum	T1	Fuel	1600	292	109	500	700
Petroleum	T1	Fuel	1800	350	135	777	850
Petroleum	T1	Fuel	1953	419	150	847	953
Diesel	T1	Fuel	25	177	100	490	460
Diesel	T1	Fuel	31	300	150	600	520
Diesel	T1	Fuel	47	444	220	700	645
Diesel	T1	Fuel	66	602	275	746	700
Kerosene	T1	Fuel	35	280	150	534	556
Kerosene	T1	Fuel	71	320	170	600	600
Kerosene	T1	Fuel	95	400	222	750	650
Kerosene	T1	Fuel	122	546	294	1023	730
Methanol	T2	VOC	1566	3935	599	2283	1031
Methanol	T2	VOC	2081	4275	690	2460	1272
Methanol	T2	VOC	2918	4523	775	2605	1440
Methanol	T2	VOC	4079	4805	872	2760	1640
Acetone	T2	VOC	8029	812	100	700	900
Acetone	T2	VOC	11974	1000	220	800	1050
Acetone	T2	VOC	15919	1155	292	871	1160
Acetone	T2	VOC	23809	1280	340	1020	1362
CE	T3	Simulant	2	730	280	1250	1200
CE	T3	Simulant	20	895	360	1660	1615

CE	T3	Simulant	35	1430	385	2015	1770
CE	T3	Simulant	44	1480	480	2120	2035
DE	T3	Simulant	68	121	393	806	417
DE	T3	Simulant	80	218	450	1169	585
DE	T3	Simulant	94	278	530	1223	665
DE	T3	Simulant	111	330	681	1400	754
DB	T3	Simulant	28	150	100	800	510
DB	T3	Simulant	56	315	166	950	900
DB	T3	Simulant	79	370	210	1085	1040
DB	T3	Simulant	116	528	233	1230	1195

3.3 Principal Component Analysis (PCA)

It was applied to standardize the sensor data to reduce dimensionality, analyze the sensor redundancy and visualize gas separation. The covariance matrix of the sensor data was calculated. The maximum variance was retained by the first three principal components.

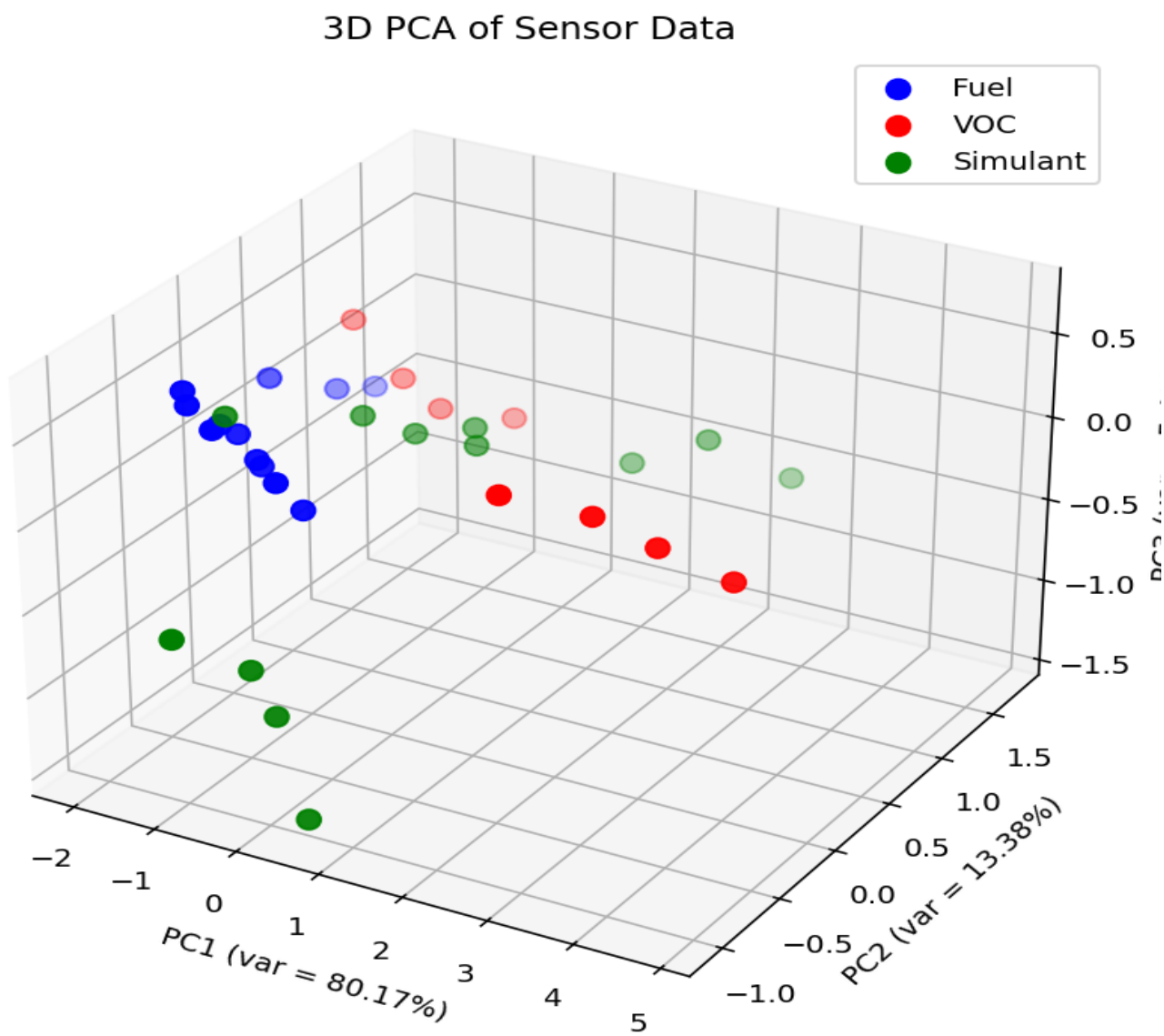


Figure 2 a View 1

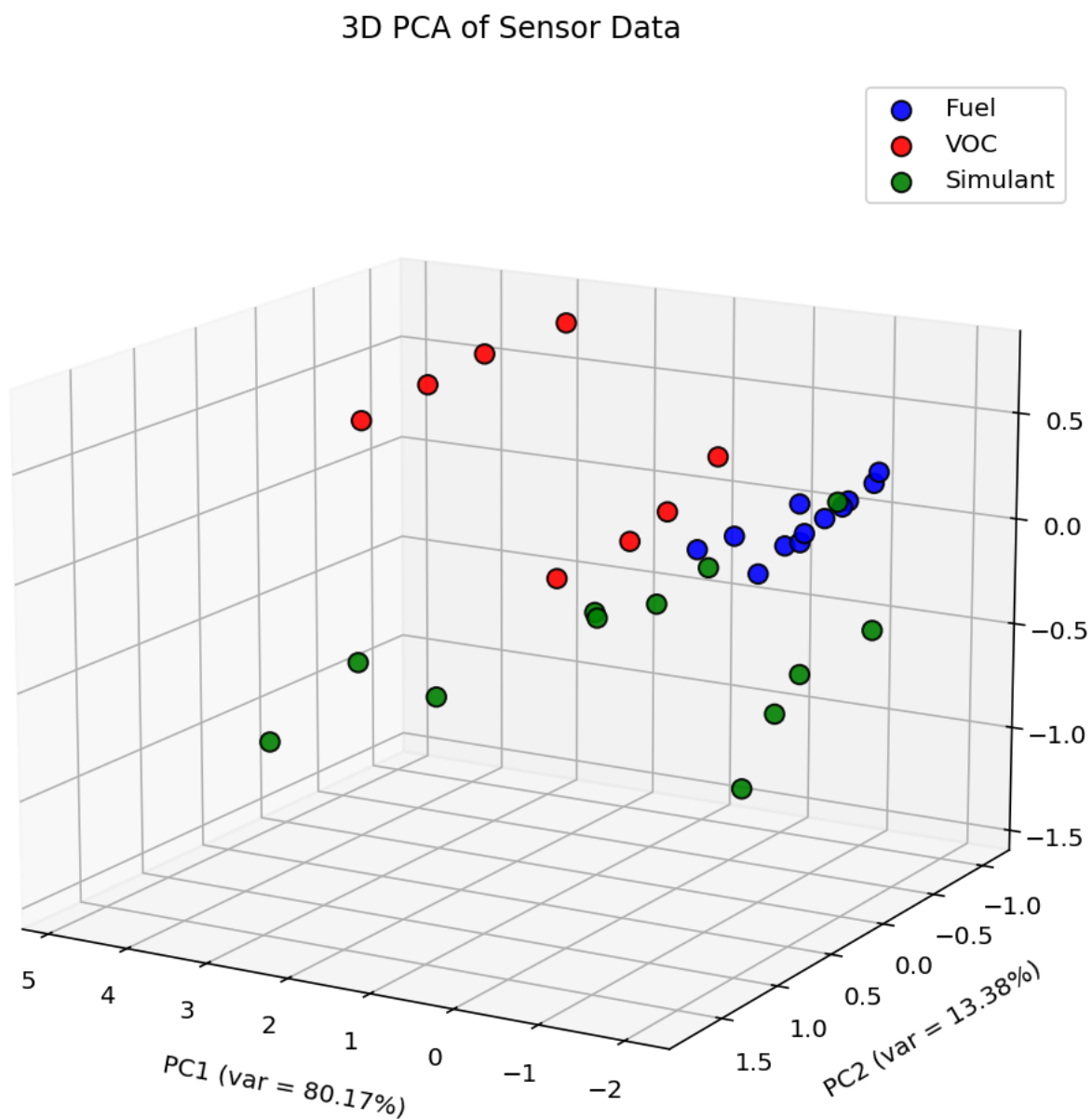


Figure 2 b View 2

Figure 2: 3D PCA Score Plot of E-Nose Sensor Data

The 3D score plot visually examines the clustering of different target analytes and overlapping classes.

3.4 Redundancy Analysis using PCA

The redundancy among sensors was identified by analyzing the contribution of each sensors to the principal component. Sensors which exhibit same loading patterns were considered to be correlated.

Table 2 Variance Contribution of Principal Components

Principal Component	Sensor1	Sensor2	Sensor3	Sensor4
PC1	0.508	0.499	0.550	0.437
PC2	-0.255	-0.471	-0.009	0.845
PC3	0.788	-0.598	-0.110	-0.097

This justifies the dimensionality reduction.

3.5 ANN using Raw Sensor Data

The raw sensor data was used as input to create an Artificial Neural Network (ANN). Neurons corresponding to the individual gas sensors made the input layer. Corresponding to the target analytes the network architecture consisted of an input layer, one or more hidden layers, and an output layer. Multiple experiments were performed by varying the number of hidden neurons, number of hidden layers and transfer functions.

The network was trained using backpropagation algorithm and the performance was evaluated using the RMS error.

3.6 Integrated PCA-ANN Approach

In this approach, a reduced set of principal components were obtained by applying the raw data to PCA. These principal components were then used as inputs for the ANN. It uses the same learning algorithm for training as the raw ANN model. This approach reduces input dimensionality and improves the learning efficiency.

3.6 Training, Testing and Performance Evaluation

Two subsets are formed by dividing the data set - Training and Testing. The training dataset was used to adjust network weights, and the testing data was used to evaluate the capacity of generalization of the trained ANN and PCA-ANN models. The Root Mean Square (RMS) error was used to evaluate the performance of the models.

2. Results and Discussion

The results obtained from the E-nose based gas detection system using PCA, ANN, and PCA-ANN are presented below.

4.1 Target Analytes and Sensor Data

When exposed to the E-nose sensor array the selected target analytes showed clear and consistent response. This confirms their suitability for the classification using pattern recognition-based techniques. Refer Table 1

4.2 PCA Score Plot Results

The plot given in figure 1 reveals that the response signature for different analytes was distinct. Majority of the variance was retained by the first three principal components which results in clear separation between gas classes in reduced dimensional space. Effective dimensionality reduction and reduced sensor redundancy is identified by the minimal overlap between the clusters. This confirms that PCA successfully extracts distinguishing features and give suitable inputs for the ANN classification.

4.3 ANN Training

In the ANN training using raw data we observed that the RMS error varies with the number of hidden neurons. When a smaller number of hidden neurons are used, a higher RMS error results indicating that the learning performance of the network has been limited. Increasing the number of hidden neurons results in progressively declining RMS errors until they reach a minimum at 12 hidden neurons. Beyond that point, an increase in RMS error did occur but showed that no further improvement in performance would result. Thus, an ANN containing 12 hidden neurons provides optimal performance for gas detection.

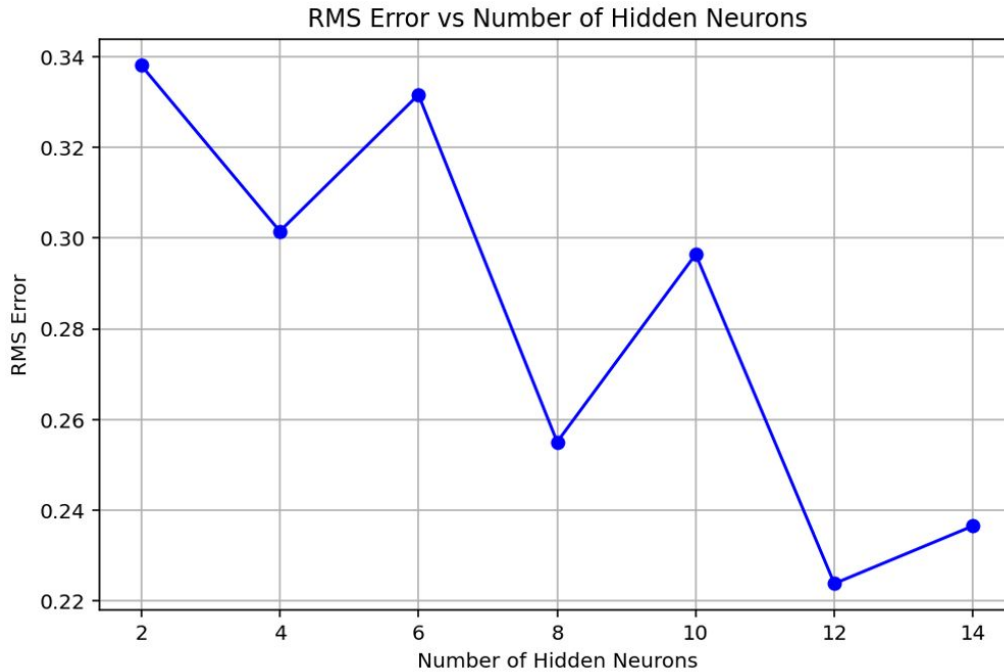


Figure 3 RMS Error Plot During ANN Training

4.4 PCA-ANN Training

In the ANN training using PCA-reduced sensor data we observed that the RMS error varies with the number of hidden neurons. A larger RMS error value is achieved with a smaller number of hidden neurons, which signifies less learning even after applying dimensionality reduction. When there are more neurons in the hidden layer, the RMS errors go down until there is an optimum at 10 hidden neurons. After that, the RMS error increases ever so slightly indicating there is no longer improved performance. The model of PCA-ANN with 10 hidden neurons represents the best compromise between prediction ability and model complexity.

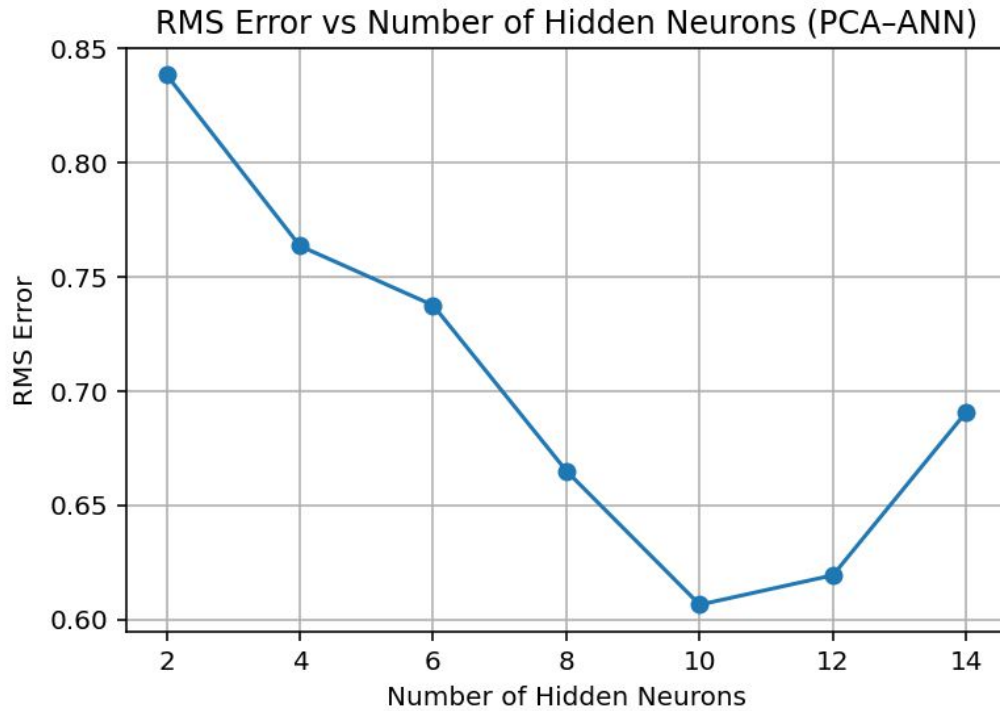


Figure 4 RMS Error Plot During PCA-ANN Training

The performance of PCA-ANN and ANN model was assessed using RMS error as a function of the number of hidden neurons. The ANN model trained using raw sensor data successfully achieved lower RMS error, however it required a greater number of neurons. The PCA-ANN model achieved perfect performance with lesser number of neurons because of the reduced input dimensionality. Hence, for practical implementations where computational efficiency and robustness is critical, PCA-ANN model is preferred.

4.5 ANN Testing

Reasonable classification performance was seen when the trained ANN model was tested with the unseen data. However, limitations due to redundant and correlated sensor inputs are indicated by the relatively higher RMS error.

Table 3 ANN Testing Output

Vapors	Target label	Category	Predicted Label	Match
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T1	TRUE
Petroleum	T1	Fuel	T3	FALSE
Diesel	T1	Fuel	T1	TRUE
Diesel	T1	Fuel	T1	TRUE
Diesel	T1	Fuel	T1	TRUE
Diesel	T1	Fuel	T1	TRUE
Diesel	T1	Fuel	T1	TRUE
Diesel	T1	Fuel	T1	TRUE
Diesel	T1	Fuel	T1	TRUE
Diesel	T1	Fuel	T1	TRUE
Diesel	T1	Fuel	T1	TRUE
Diesel	T1	Fuel	T1	TRUE
Diesel	T1	Fuel	T1	TRUE
Kerosene	T1	Fuel	T1	TRUE
Kerosene	T1	Fuel	T1	TRUE
Kerosene	T1	Fuel	T1	TRUE

Kerosene	T1	Fuel	T1	TRUE
Kerosene	T1	Fuel	T1	TRUE
Kerosene	T1	Fuel	T1	TRUE
Kerosene	T1	Fuel	T1	TRUE
Methanol	T2	VOC	T2	TRUE
Methanol	T2	VOC	T2	TRUE
Methanol	T2	VOC	T2	TRUE
Methanol	T2	VOC	T2	TRUE
Methanol	T2	VOC	T2	TRUE
Methanol	T2	VOC	T2	TRUE
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DE	T3	Simulant	T1	FALSE
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DB	T3	Simulant	T3	TRUE
DB	T3	Simulant	T3	TRUE
DB	T3	Simulant	T3	TRUE
DB	T3	Simulant	T3	TRUE

Table 3 presents the ANN testing output using the raw sensor data. In this table the target labels are being compared with the predicted labels which are generated by the trained ANN model.

Majority of the testing sample were correctly classified, indicated by TRUE match status. It demonstrates the effective pattern recognition capability of the network. The high classification accuracy observed during testing is consistent with the low RMS error obtained during ANN training.

4.6 PCA-ANN Testing

The PCA-ANN model showed better testing performance with lower RMS error and better generalization capability. Hence it is confirmed that the integration of PCA before ANN classification is more effective.

Table 4 PCA-ANN Testing Output

PC1	PC2	PC3	Target label	Out1	Out2	Out3	Pred_Label	Match
-1.45857	0.265008	0.272979	T1	0.969298	0.00012	0.030582	T1	TRUE
-1.32348	0.234564	0.245073	T1	0.962518	0.000221	0.037261	T1	TRUE
-0.69616	-0.0225	-0.4172	T1	0.570605	0.190745	0.23865	T1	TRUE
-1.37779	0.138905	-0.08829	T1	0.955979	0.000914	0.043108	T1	TRUE
-1.15832	0.208182	0.264377	T1	0.954094	0.000352	0.045554	T1	TRUE
-0.49883	0.173808	0.302792	T1	0.618669	0.006272	0.37506	T1	TRUE
-1.76583	0.393437	0.320798	T1	0.979483	4.02E-05	0.020476	T1	TRUE
-0.85233	0.031185	-0.32582	T1	0.814859	0.051295	0.133846	T1	TRUE

-1.56944	0.233417	0.057789	T1	0.968986	0.000245	0.030769	T1	TRUE
-1.51397	0.363165	0.334219	T1	0.971354	9.36E-05	0.028552	T1	TRUE
-1.14416	0.24713	0.296794	T1	0.953296	0.000352	0.046352	T1	TRUE
-1.25929	0.121222	-0.13665	T1	0.945511	0.001888	0.052601	T1	TRUE
-1.61188	0.379402	0.327193	T1	0.974719	6.92E-05	0.025212	T1	TRUE
-0.60925	0.192865	-0.45479	T2	0.344947	0.510992	0.144061	T2	TRUE
4.031445	1.419565	0.115866	T2	2.8E-05	0.995715	0.004257	T2	TRUE
0.285537	-0.14511	-0.51553	T2	0.01421	0.493116	0.492674	T2	TRUE
3.459871	1.563285	0.206392	T2	6.61E-05	0.993784	0.00615	T2	TRUE
0.631556	-0.28095	-0.69235	T2	0.004235	0.34357	0.652195	T3	FALSE
4.525141	1.279214	0.042319	T2	7.98E-06	0.997397	0.002596	T2	TRUE
-0.40311	-0.00512	0.945772	T3	0.003655	2.47E-05	0.99632	T3	TRUE
1.156457	-1.3159	0.434723	T3	3.1E-09	2.72E-08	1	T3	TRUE
3.817304	-2.97796	0.873917	T3	1.75E-19	3.16E-15	1	T3	TRUE
-0.07274	-1.01461	0.004964	T3	5.28E-06	8.45E-07	0.999994	T3	TRUE
-0.15038	-0.52505	-0.08683	T3	0.000978	0.000286	0.998736	T3	TRUE
1.396514	-0.60939	-1.11259	T3	0.000189	0.105141	0.894669	T3	TRUE
-0.097	-0.17102	1.004399	T3	0.00028	6.91E-06	0.999713	T3	TRUE
1.483543	-1.46932	0.359184	T3	4.06E-10	2.52E-08	1	T3	TRUE
0.38826	-1.27559	0.095248	T3	9.54E-08	9.43E-08	1	T3	TRUE

Table 4 Shows the testing results of the integrated PCA-ANN model. The ANN inputs are comprised of the three principal components PC1, PC2, and PC3 provided from PCA. The ANN outputs (Out1, Out2 and Out3) will provide the expected rear probabilities of each of the three target classes. The maximum output value is used to determine the predicted label.

The coupling of PCA and ANN is a major factor to the greatly enhanced classification capability of the E_{nose} system. PCA eliminates redundant data and noise, thus by mapping the original sensor data into a small set of uncorrelated features, it develops new features that are more informative and easier for ANN to learn. Consequently, the PCA-ANN model has been demonstrated to produce the better predictive performance with lower RMS error and higher testing accuracy than the ANN model trained on the raw sensor data.

Table 5 Comparison in ANN and PCA-ANN approach

Model	Input Features	Accuracy	Training Time	Overfitting	Stability
ANN without PCA	High-dimensional	Lower	Higher	More	Less
ANN with PCA	Reduced	Higher	Lower	Less	More

3. Conclusion

The current project has successfully illustrated the use of AI-based methods for gas sensing using an electronic nose system. PCA achieved a successful reduction in the number of dimensions in the sensor's Array while eliminating redundant data. The Artificial Neural Network performed better in terms of learning efficiency and classification when trained on the PCA-reduced features. Compared to the PCA and ANN models independently, the PCA-ANN integrated model was faster and more accurate. This paper demonstrates how statistical models and machine learning models can be used to solve scientific problems with great success. The approach can be generalized to more complex gas sensing problems in the future.

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